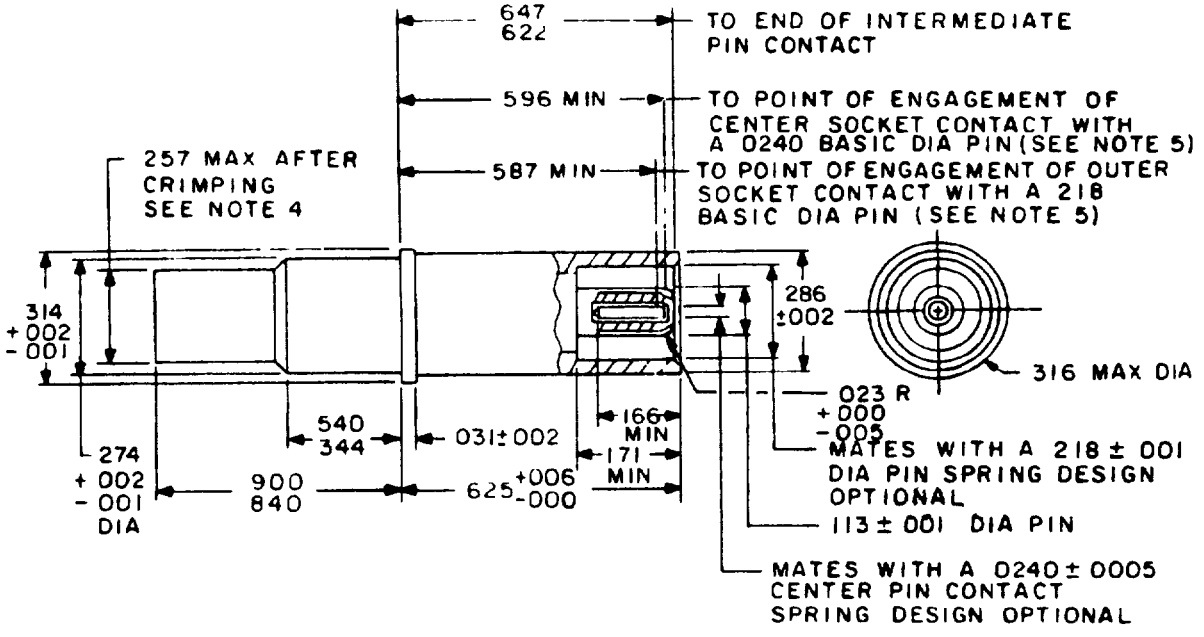
MILITARY SPECIFICATION SHEET

# CONTACT, ELECTRICAL CONNECTOR, CONCENTRIC TWINAX, SOCKET, SHEILDED, SIZE 8

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the contact described herein shall consist of this specification and the latest issue of MIL-C-39029.




| INCHES | MM | INCHES | MM | INCHES | MM | INCHES | MM |
|---|---|---|---|---|---|---|---|
| .0005 | 0.013 | .0240 | 0.610 | .257 | 6.53 | .540 | 13.72 |
| .001 | 0.03 | .031 | 0.79 | .274 | 6.96 | .587 | 14.91 |
| .002 | 0.05 | .113 | 2.78 | .286 | 7.26 | .596 | 15.14 |
| .005 | 0.13 | .166 | 2.95 | .314 | 7.98 | .622 | 15.80 |
| .006 | 0.15 | .171 | 4.34 | .316 | 8.03 | .625 | 15.88 |
| .023 | 0.58 | .218 | 5.54 | .344 | 8.74 | .647 | 16.43 |
| | | | | | | .840 | 21.34 |
| | | | | | | .900 | 22.86 |

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.
4. Diameter shall not exceed .276 (7.01 mm) over recovered heat shrink tubing.
5. Point at which a square ended pin of the same basic diameter as the mating contact spring. Provisions for clearance hole shall be provided for outer contact test pin.

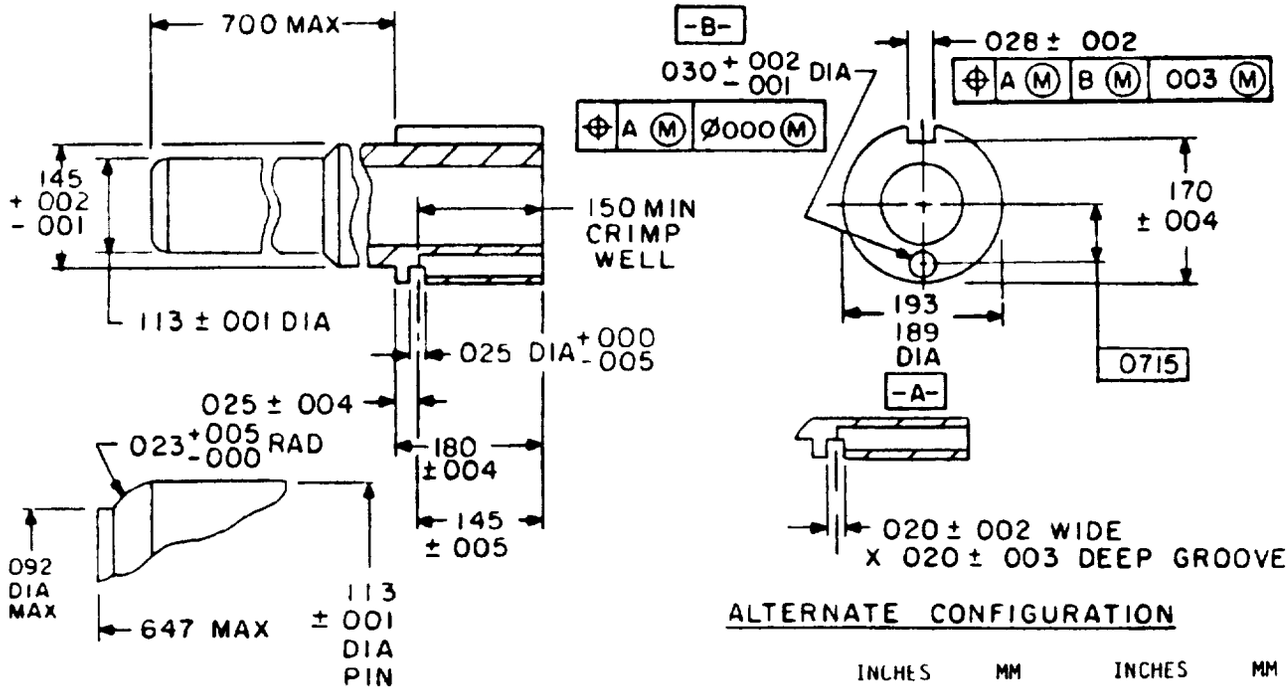
Ⓐ FIGURE 1. Concentric twinax contact.

Ⓐ denotes changes





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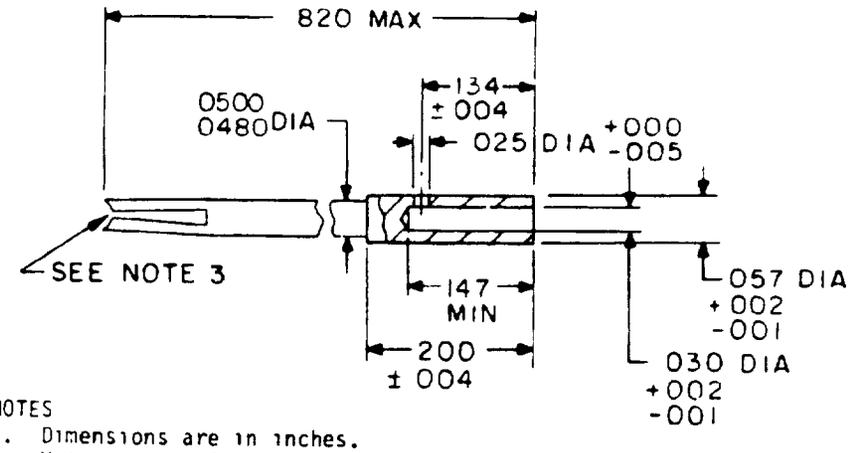

ALTERNATE CONFIGURATION

INTERMEDIATE PIN END ALTERNATE DESIGN

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

| INCHES | MM | INCHES | MM |
|---|---|---|---|
| .001 | 0.03 | .092 | 2.34 |
| .002 | 0.05 | .113 | 2.87 |
| .003 | 0.08 | .145 | 3.68 |
| .004 | 0.10 | .150 | 3.81 |
| .005 | 0.13 | .170 | 4.32 |
| .020 | 0.51 | .180 | 4.57 |
| .023 | 0.58 | .189 | 4.80 |
| .025 | 0.64 | .193 | 4.90 |
| .028 | 0.71 | .647 | 16.43 |
| .0715 | 1.816 | .700 | 17.78 |

Ⓐ FIGURE 2. Intermediate pin contact.



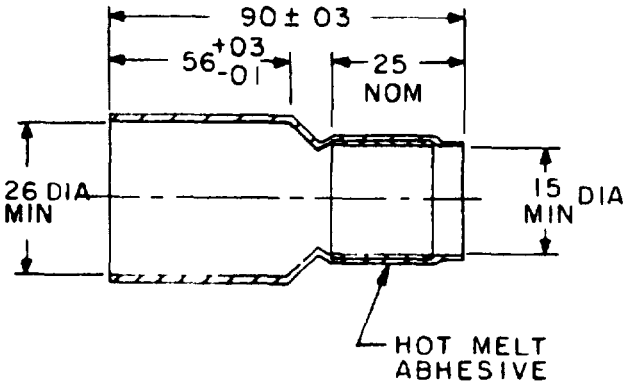

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Socket must mate with 0 0240+ 0005 pin Closed entry may be provided either by an integral part of contact design or by the surrounding dielectric material

| INCHES | MM |
|---|---|
| .001 | 0.03 |
| .002 | 0.05 |
| .004 | 0.10 |
| .005 | 0.13 |
| .025 | 0.64 |
| .030 | 0.76 |
| .0480 | 1.220 |
| .0500 | 1.270 |
| .057 | 1.45 |
| .134 | 3.40 |
| .147 | 3.73 |
| .200 | 5.08 |
| .820 | 20.83 |

Ⓐ FIGURE 3. Center socket contact.



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| INCHES | MM |
|---|---|
| .01 | 0.3 |
| .03 | 0.8 |
| .15 | 3.8 |
| .25 | 6.4 |
| .26 | 6.6 |
| .56 | 14.2 |
| .90 | 22.9 |

NOTES
1. Dimension are in inches.
2. Metric equivalents are given for general information only.

Ⓐ FIGURE 4. Heat shrink boot.



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REQUIREMENTS

(A) Qualification Contacts shall comply with reliability assurance provisions of MIL-STD-790 as specified in MIL-C-38999.

Design and construction

Dimensions and configuration See figures 1 through 4 and table I

TABLE I. Design characteristics.

| BIN code | Color bands 1st | 2nd | 3rd | Contact cavity size | Cable accommodated | Type | Class |
|---|---|---|---|---|---|---|---|
| (A) 530 | Green | Orange | Black | 9 | M17/176-00002 | D | B |

Assembly procedure Manufacturer's recommended assembly instructions shall be shipped with unit package.

(A) Material

Heat shrink boot Shall be in accordance with MIL-I-23053/8.

Electrical

Low signal level contact resistance (center and intermediate contacts only) See table II

Contact resistance See table III.

Frequency 0 to 20 MHz (operating frequency range).

Voltage rating 500 volts rms maximum, working voltage at sea level, 125 volts rms maximum at 70,000 feet.

TABLE II. Low signal level contact resistance (center and intermediate contacts only)

| BIN code | Cable accommodated | Maximum contact resistance (milliohms) Initial | After conditioning |
|---|---|---|---|
| (A) 530 | M17/176-00002 | 55 | 66 |

TABLE III. Contact resistance.

| BIN code | Contact | Cable accommodated | Test current (amperes) | Maximum voltage drop (millivolts) 25°C, +3°C, -0°C Initial | 25°C, +3°C, -0°C After conditioning | +175°C, +3°C, -0°C After conditioning |
|---|---|---|---|---|---|---|
| (A) 530 | Center | M17/176-00002 | 1.0 | 55 | 66 | 94 |
| 530 | Intermediate | M17/176-00002 | 1.0 | 55 | 66 | 94 |
| 530 | Outer | M17/176-00002 | 12.0 | 75 | 90 | 129 |



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Dielectric withstanding voltage  Shall be as specified in table IV.

TABLE IV  Dielectric withstanding voltage.

| Contacts | Altitude | Test voltages ac rms |
|---|---|---|
| Center to intermediate | Sea level | 1000 |
| Intermediate to outer | Sea level | 500 |

(A) Mating contact  Shall be in accordance with MIL-C-39039/90.

Mechanical

(A) Contact engagement and separation force (socket contacts only)  The engagement depth shall be a minimum of 0.7 of the minimum socket bored.  The test pins shall be in accordance with MS3197, except the diameters shall be as specified in table V.  Provision for clearance hole on outer contact.  Test pins shall be provided.  The test pins shall be in accordance with MS3197, except the diameters shall be as specified in table V.

(A) TABLE V  Contact engagement and separation force.

| Test pin diameter (inch) | Minimum separation force (ounces) | | Maximum engagement force (ounces) | | Maximum average engagement force |
|---|---|---|---|---|---|
| | Initial | After conditioning | Initial | After conditioning | |
| .2190 +.0000 -.0001 | NA | NA | 48 | 60 | NA |
| .2170 +.0001 -.0000 | 3.0 | 2.0 | NA | NA | NA |
| .0245 +.0000 -.0001 | NA | NA | 12 | 14 | NA |
| .0235 +.0001 -.0000 | 0.5 | 0.4 | NA | NA | NA |

Crimp tensile strength (center, intermediate, and outer contact crimp joint).  See table VI.

(A) TABLE VI.  Crimp tensile strength (at ambient).

| BIN code | Cable accommodated | Axial load (pounds, minimum) | | |
|---|---|---|---|---|
| | | Center contact | Intermediate contact | Outer contact |
| 530 | M17/176-00002 | 9 | 8 | 25 |



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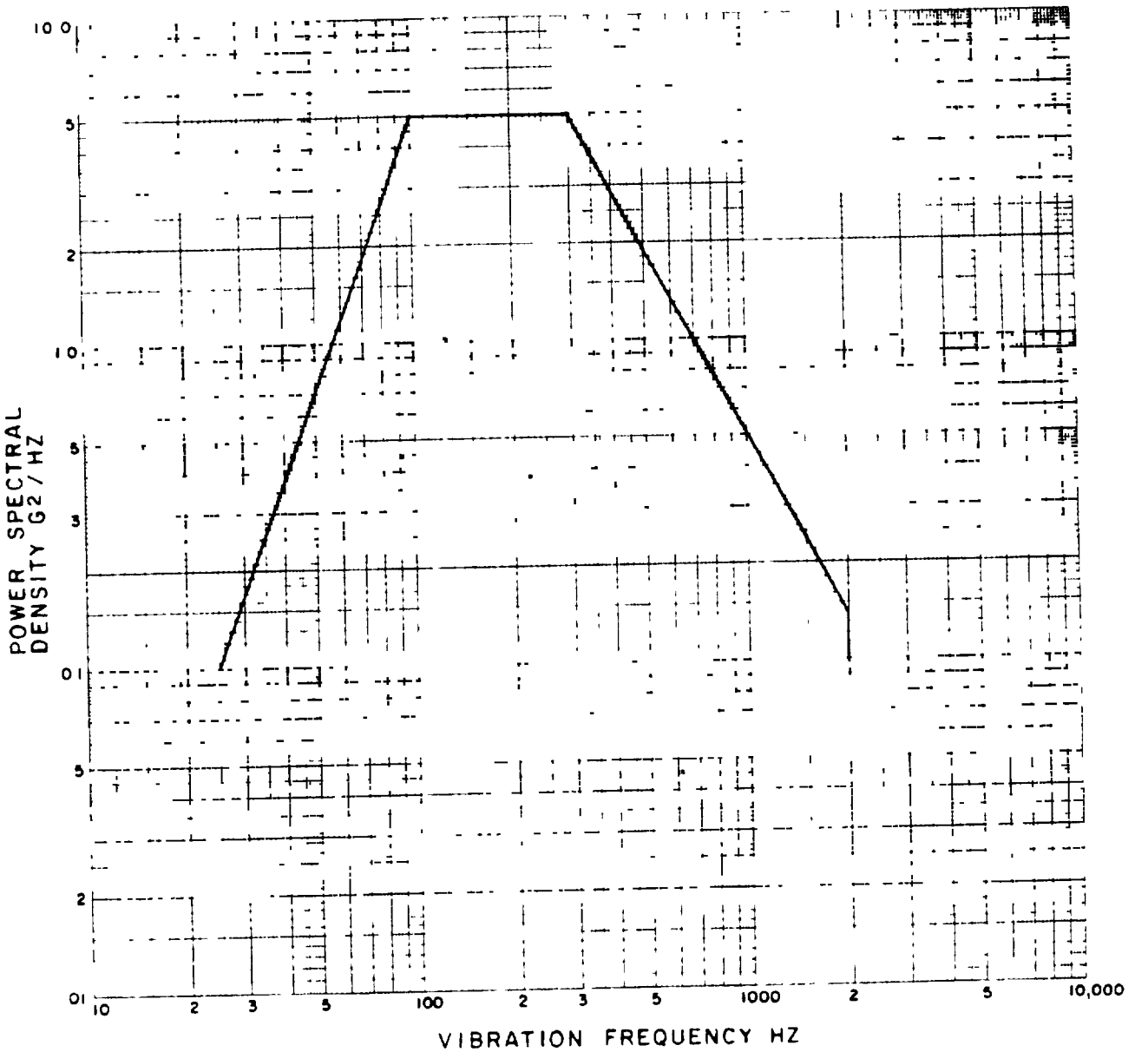



FIGURE 5. Vibration envelope.



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Vibration Method 2005 of MIL-STD-1344, test condition V. The following details shall apply.

a. Use the vibration envelope shown on figure 5.

b. Vibration To be conducted at standard test conditions.

c. Duration 8 hours in the longitudinal direction and 8 hours in a perpendicular direction (16 hours).

High impact shock All contacts shall be wired in a series circuit with 100 milliamperes maximum current flow through the series circuit during shock. Connectors shall be monitored for any discontinuities. A detector capable of detecting all discontinuities in excess of 1 microsecond shall be used. Wired and mated connectors shall be subjected to the test specified in MIL-S-901, grade A with the following modifications and additions. Mounting fixture shall be in accordance with MIL-S-901, lightweight The cable or wire bundle shall be supported on a stationary frame to provide a free flexing cable length between the frame and fixture of not less than 36 inches (914.4 mm).

a Coupling Coupled by normal mounting means.

b. Test condition A. Plug shall be terminated with at least 80 percent of wired contacts. The wire bundle shall be provided with straight, open frame, strain relief accessory hardware.

Tools Shall be as specified in tables VII and VIII.

(A)

TABLE VII. Tools (installing and removal).

| BIN code | Installing and removal tool | Removal tool |
|---|---|---|
| 530 | M81969/14-06 | M81969/14-06 |
| 530 | M81969/46-06 | M81969/46-12 |

(A)

TABLE VIII. Tools (crimping).

| BIN code | Center contact tooling | | Intermediate contact tooling | | Outer crimp sleeve tooling | |
|---|---|---|---|---|---|---|
| | Basic crimping tool | Contact positioner 1/ | Basic crimping tool | Die part no 1/ | Basic crimping tool | Die part no 1/ |
| 529 | M22520/2-01 | K709 | M22520/5-01 | Y631 Die closure B | M22520/5-01 | Y631 Die closure A |

Environmental

Temperature rating $-65^{\circ}C$ to $+175^{\circ}C$

(A) Humidity Perform humidity upon completion of group 1, using qualified MIL-C-38999H connectors with the 25-20 insert arrangment. When tested as specified below, wired, mated contacts shall show no deterioration which will adversely affect the performance of the contact. Following the test and during the final cycle, insulation resistance shall exceed 100 megohms at 25°C. Insulation resistance shall be measured between the crimped outer and intermediate contacts and between the crimped inner and intermediate contacts in accordance with method 3003 of MIL-STD-1344.

1/ Daniels Manufacturing Corporation part number is specified Tools may be provided by other vendors if they are determined to be equivalent by the Qualifying Activity



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Humidity test: Wired, mated connectors shall be subjected to the humidity test specified in method 1002 of MIL-STD-1344. The following details and exceptions shall apply

a. Test condition - Type II omitting subcycle step 7 (b).

b. The mated connectors shall be mounted in a vertical position.

c Step 7 (a) shall be performed during the last cycle.

d. Three hours minimum after the start of step 7 (a) during the final cycle and while the connectors are still subjected to high humidity, the insulation resistance shall be measured when the chamber temperature reaches 20°C ±5°C and condensation is observed on the connector.

Part number See table IX.

TABLE IX. Part number.

| BIN code | Part number | Supersedes |
|---|---|---|
| 530 | M39029/91-530 | --- |

QPL evaluating activity Naval Avionics Center, D/714, 6000 East 21st Street, Indianapolis, Indiana 46218.

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities
Navy - EC, SH, MC
Air Force - 11, 15, 17, 99
DLA - ES

User activities
Army - AT
Navy - OS

Agent
DLA - ES

Preparing activity
Navy - AS

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